



SS1 PHYSICS LESSON NOTES (THIRD TERM)
WEEK 2:

LESSON NOTE: THIRD TERM - SS2 WEEK 2

Subject: Chemistry / Physics (Electrochemistry)

Class: SS2

Term: Third Term

Week: 2

Topic: Electroplating

1. DEFINITION OF ELECTROPLATING

Electroplating is the process of depositing a thin layer of one metal onto the surface of another metal object using electric current and an electrolyte solution.

It is a type of electrolysis used mainly for:

Coating

Protection

Decoration



Improving conductivity of objects

2. BASIC PRINCIPLE OF ELECTROPLATING

Based on **electrolysis**, where direct current (DC) passes through an electrolyte solution containing metal ions.

The metal to be deposited is connected as the **anode (+)**, and the object to be coated is the **cathode (-)**

When electricity flows, positive metal ions from the electrolyte are **reduced at the cathode** and form a coating.

Meanwhile, the **anode dissolves** to replenish metal ions in solution (if it is made of the plating metal)

3. COMPONENTS OF AN ELECTROPLATING SYSTEM

Component	Description
Power Supply	Provides direct current (DC) to drive the reaction
Cathode (-)	The object to be plated (e.g., steel spoon)
Anode (+)	Metal used for plating (e.g., copper, nickel)
Electrolyte	A solution containing ions of the plating metal (e.g., CuSO_4)



Component	Description
Connecting wires	To connect the electrodes to the power supply

4. ELECTROCHEMICAL REACTIONS IN ELECTROPLATING

Example: Electroplating a spoon with copper

Electrolyte: Copper(II) sulfate solution (CuSO_4)

Anode: Copper plate

Cathode: Iron or steel spoon

Electrode	Reaction
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At Cathode (-)	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu(s)}$ (Copper ions are reduced and deposited as metallic copper on the spoon)
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At Anode (+)	$\text{Cu(s)} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ (Copper from the anode dissolves to replace ions in the solution)
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Note: The thickness of the deposited metal depends on current, time, concentration, and temperature

5. GENERAL CONDITIONS FOR EFFECTIVE ELECTROPLATING

Clean surface of the object to be plated

Constant DC current supply

Correct concentration of electrolyte

Adequate temperature (moderately warm is ideal)

Proper electrode placement

Duration long enough for desired thickness

6. COMMON METALS USED FOR ELECTROPLATING

Plating Metal

Reason

Nickel

Corrosion resistance and aesthetic shine

Chrome

Hardness, corrosion resistance

Gold/Silver

Decorative, conductivity (electronics)

Zinc

Protects steel from rust (galvanization)



Plating Metal

Reason

Copper

Base layer for other plating or aesthetics

7. APPLICATIONS OF ELECTROPLATING

Jewelry coating with gold or silver

Automobile parts (bumpers, handles)

Cutlery and kitchenware

Electrical connectors and circuit board

Coins and medals

Protective coating on tools and machine

Galvanization (zinc-plating of iron to prevent rust)

8. ADVANTAGES OF ELECTROPLATING

Enhances appearance (shiny, decorative)



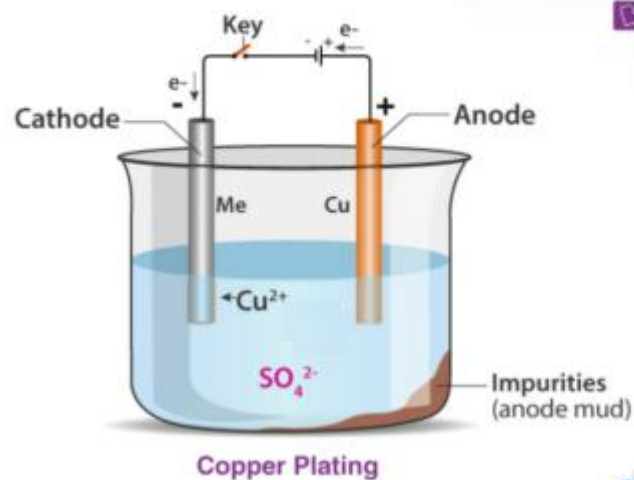
- Prevents corrosion and rust
- Improves electrical conductivity
- Increases surface hardness and durability
- Reduces friction in machinery parts
- Makes cheaper metals appear valuable (e.g., gold-plated items)

9. DISADVANTAGES / LIMITATIONS OF ELECTROPLATING

- Costly if precious metals (e.g., gold, silver) are used
- Toxic chemicals used can harm environment if not properly managed
- Coated layer may wear off over time
- Requires controlled conditions for quality results
- Improper cleaning of surfaces leads to poor adhesion

10. ELECTROPLATING SETUP DIAGRAM

Electroplating Setup Diagram:



Typical setup includes:

Beaker containing electrolyte solution

Copper anode (+) connected to positive terminal

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Spoon (cathode, -) connected to negative terminal

DC power supply

Wires with crocodile clips

11. SAFETY PRECAUTIONS IN ELECTROPLATING

Use gloves and goggles to handle chemicals.

Work in a well-ventilated area.

Avoid direct skin contact with electrolyte.

Do not allow electrodes to touch inside the beaker.

Wash hands and materials after the experiment

Store chemicals safely and label properly.

Dispose of used solutions responsibly (do not pour into drains)

12. EVALUATION

Objective Questions (WAEC/NECO Style)

In electroplating, the object to be plated is connected to the:

- A. Positive terminal
- B. Negative terminal ✓
- C. Earth
- D. Neutral wire



Electroplating involves:

- A. Magnetic induction
- B. Chemical bonding
- C. Electrolysis ✓
- D. Radiation transfer

Theory Questions:

Define electroplating and explain its working principle

Draw and label a simple electroplating setup.

State four uses of electroplating in the industry

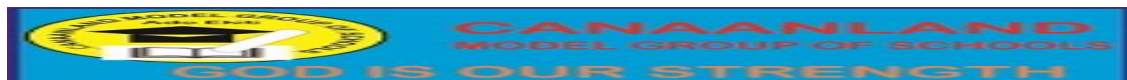
What are the reactions at the anode and cathode during copper electroplating?

List three factors that influence the quality of electroplating.

State two advantages and two disadvantages of electroplating.

13. ASSIGNMENT

- (a) Research and describe the industrial process of electroplating with chromium for car parts.
- (b) Explain how electroplating is used in preventing rust in steel bridges.



(c) Describe an experiment to electroplate an iron nail with zinc. Include materials and procedure.

